

1 Short questions

1. Piping is the mechanism in Unix that allows a process to redirect its output to a file, instead of the standard output.

Your answer(True/False) : False

2. Name the environment variable that stores the list of directories used by the shell to search for binary files to be executed

Your answer : PATH

3. The standard error stream *stderr* can be used by the Unix system calls *read()* and *write()*.

Your answer(True/False) : False

4. The function **fgets()** reads characters from a stream into an array and adds a new line character at the end.

Your answer(True/False) : False

5. What is the difference between unnamed pipes and FIFOs?

Your answer : (1) FIFOs exist in the file system while pipes don't and, (2) unnamed pipes are used by related processes while FIFOs can be used between any processes within a Unix system.

6. What is the difference between sockets and FIFOs?

Your answer : FIFOs are local to a single machine, whereas sockets can be used across any two machines.

7. A process can block or ignore a SIGKILL signal.

Your answer(True/False) : False

8. When a process **P** creates a child process **P1**, then creates another child process **P2**, the pid of **P1** is greater than the pid of **P2**.

Your answer : False

9. System calls **read()** and **write()** use buffering when dealing with files.

Your answer(True/False) : False

10. How many values does the system call **fork()** return if successful?

Your answer : 2

11. How many values does the system call **fork()** return if it fails?

Your answer : 1

12. Every user process must have a PID and a PPID, what happens to the PPID of a child process when its parent dies while the child is still alive?

Your answer : PPID values is changed to 1

13. When using sockets in the client/server context, a client program only needs to know the IP of the server's machine in order to get connected to the server.

Your answer (True/False): **False**

14. A socket has a name that exists in the unix file system.

Your answer (True/False): **False**

15. We can read from and write to the same FIFO.

Your answer (True/False): **False**

2 Tracing(6 questions) : 6 marks each

1. Consider the program

```
int main(int argc, char *argv[]){
    int fd;
    char buffer1[10] = "One ";
    char buffer2[10] = "Two ";

    fd = open("file.txt", O_WRONLY|O_CREAT|O_TRUNC);
    write(fd, buffer1, strlen(buffer1));
    write(STDOUT_FILENO, buffer2, strlen(buffer2));
    // STDOUT_FILENO is the screen file descriptor
    dup2(fd, STDOUT_FILENO);
    write(fd, buffer1, strlen(buffer1));
    write(STDOUT_FILENO, buffer2, strlen(buffer2));
    close(fd);
}
```

- What will appear on the screen?

Your answer(3 marks) : Two

- What will the file **file.txt** contain?

Your answer(3 marks) : One One Two

2. How many "**Hello**"s will appear on the screen when this program is executed?

```
int main(int argc, char *argv[]){
    fork();
    printf("Hello\n");
    fork();
    printf("Hello\n");
}
```

Your answer : 6

✓ (verified)

3. How many "Hello"s will appear on the screen when this program is executed?

```
int main(int argc, char *argv[]){
    int i;

    for(i=1; i<=3; i++){
        fork();
        printf("Hello\n");
    }
    printf("Hello\n");
}
```

Your answer : 22

4. What will be printed when the program below runs?

```
int main(int argc, char *argv[]){
    printf("A ");
    fork();
    printf("B\n");
    fork();
    pause();
    printf("Bye\n");
}
```

Your answer :

A B

A B

5. What will be printed by the following program

```
void action(){
    fprintf(stderr, " Thursday ");
    exit(1);
}

int main(){
    pid_t pid, st;

    if(!(pid=fork())){
        signal(SIGALRM, action);
        alarm(1);
        while(1)
            sleep(1);
    }
```

```

        fprintf(stderr, "exam ");
    }else{
        fprintf(stderr, "Final");
        wait(&st);
        fprintf(stderr, "%d\n", WEXITSTATUS(st));
    }
}

```

Hint: WEXITSTATUS(st) extracts exit status from st

Your answer : Final Thursday 1

6. What will be printed by the following program

```

void action(){
    fprintf(stderr, "OK ");
    exit(0);
}

int main(){
    pid_t pid;

    if(!(pid=fork())){
        signal(SIGUSR1, action);
        pause();
        fprintf(stderr, "First ");
    }else{
        sleep(1);
        fprintf(stderr, "Thursday ");
        kill(pid, SIGUSR1);
    }
    sleep(1);
    fprintf(stderr, " Next ");
}

```

Your answer : Thursday OK Next

7. What will appear on the screen when the program below executes?

```

int main(int argc, char *argv[]){
    int fd;
    char name1[15] = "Bonjour\n";
    char name2[15] = "Bonsoir\n";

    fd = open("file.txt", O_WRONLY|O_CREAT|O_TRUNC);
    write(fd, name1, 8);
    close(fd);
    fd = open("file.txt", O_RDONLY);
}

```

```

    dup2(fd, 0);
    read(0, name2, 8);
    write(1, name2, 8);
    exit(0);
}

```

Your answer : Bonjour

8. The program below is a simple client that reads a single message from a remote server. Explain what crucial information is missing for this client in order for it to be working correctly?

```

int main(int argc, char *argv[]){
    char message[100];
    int n, server;
    struct sockaddr_in servAdd;

    server=socket(AF_INET, SOCK_STREAM, 0);
    servAdd.sin_family = AF_INET;
    inet_pton(AF_INET, argv[1], &servAdd.sin_addr);
    connect(server, (struct sockaddr *) &servAdd, sizeof(servAdd));
    n=read(server, message, 100);
    write(1, message, n);
    exit(0);
}

```

Your answer : Port number is missing, no client/server communication can work without a valid port number